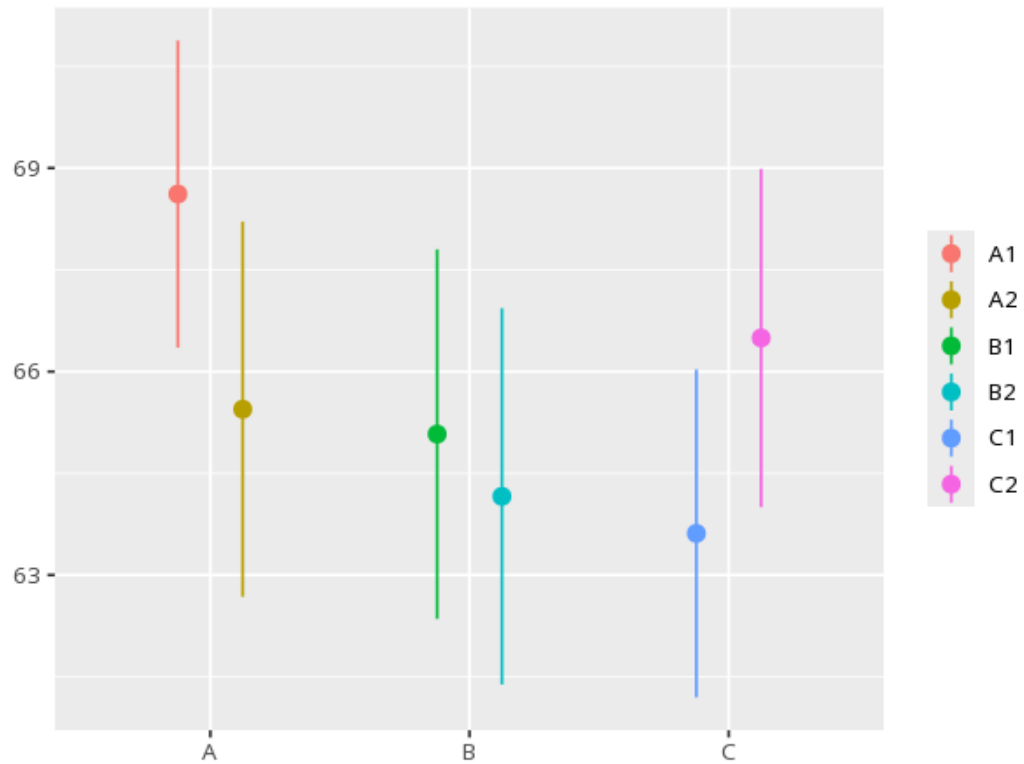


# 1 Nested ANOVA

| Group | N  | Mean  | Std. Dev. |
|-------|----|-------|-----------|
| A     | 80 | 67.03 | 8.01      |
| B     | 80 | 64.62 | 8.55      |
| C     | 80 | 65.05 | 7.77      |

| Source                       | SS     | df | MS     | F    | p    | $\omega^2$ [95% CI] |
|------------------------------|--------|----|--------|------|------|---------------------|
| school                       | 264.38 | 2  | 132.19 | 1.03 | .456 | 0.01 [0.00, 1.00]   |
| classroom (nested in school) | 384.23 | 3  | 128.08 | 1.97 | .120 | 0.01 [0.00, 1.00]   |



Used when one factor sits inside another (e.g. classes within schools). The top factor is tested against variation among its nested units, not raw residuals — a significant F means the top-level groups differ beyond the nested-unit variability. Your significance threshold ( $\alpha$ ) is 0.05.

A nested ANOVA for A gave  $F(2,3)=1.03$ ,  $p = .456$ ,  $\omega^2=.01$  [.00, 1.00].

Statistical basis: Nested ANOVA (upper factor tested against the nested-unit mean square) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.);  $\omega^2$  effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.).